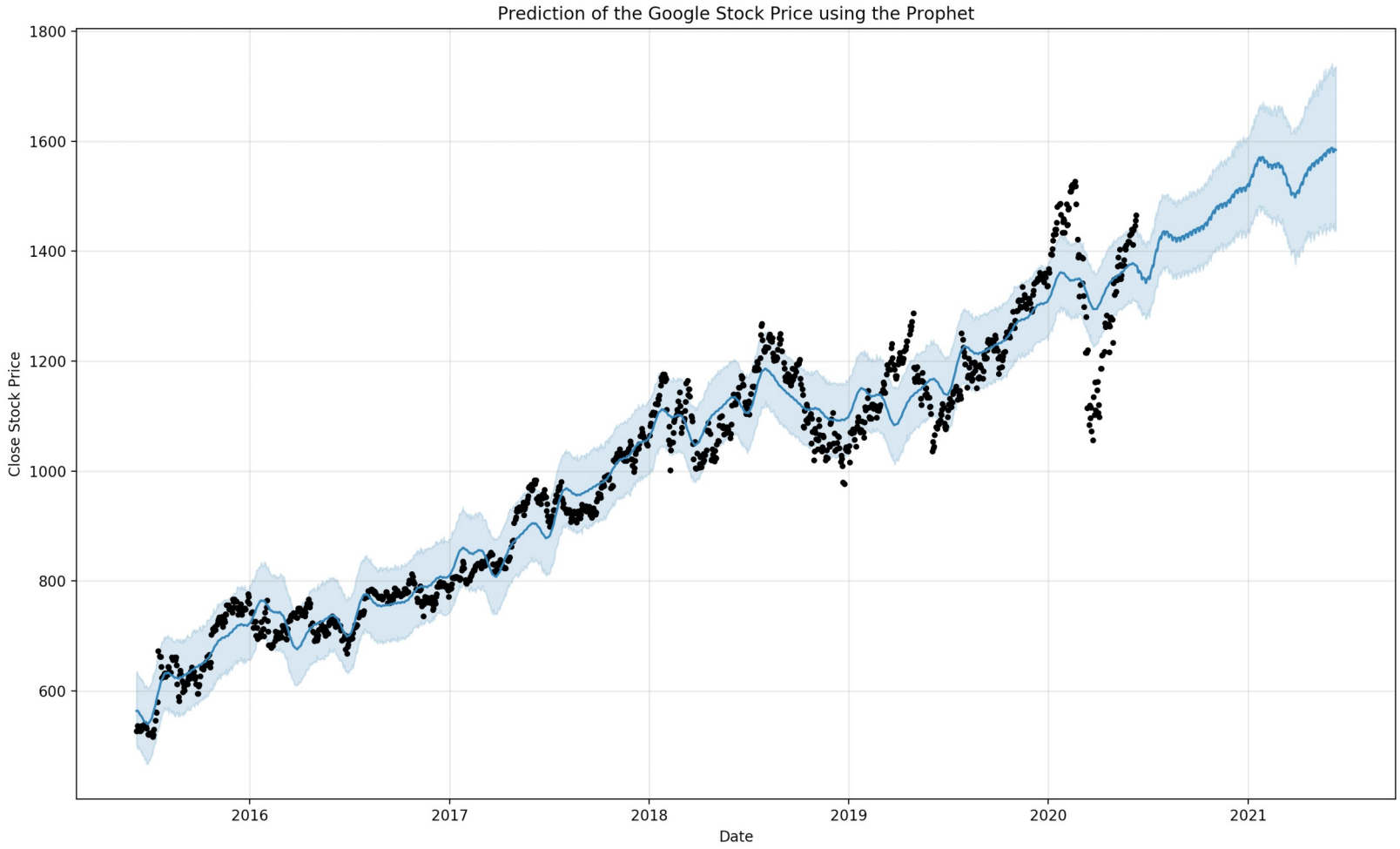
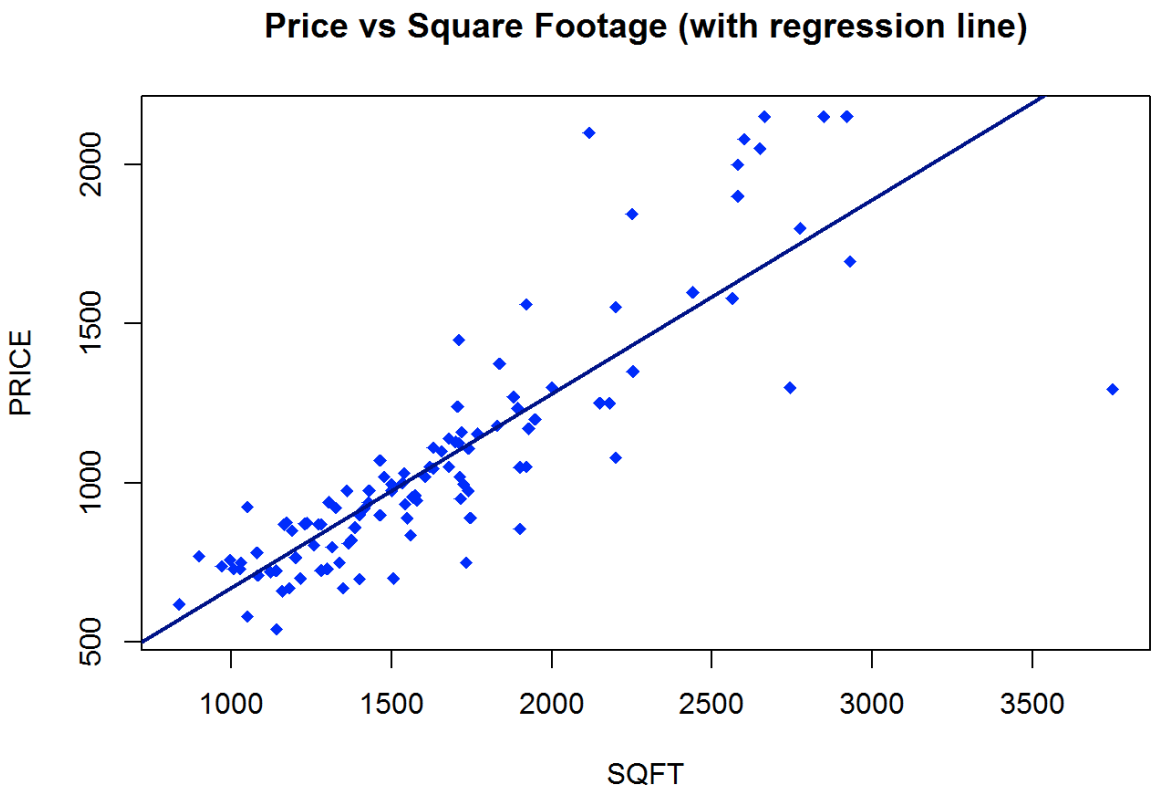
Regression


Prediction of the Google Stock Price using the Prophet
1800
1600
1400
1200
1000
800
600
Close Stock Price
2016
2017
2018
2019
2020
2021
Date

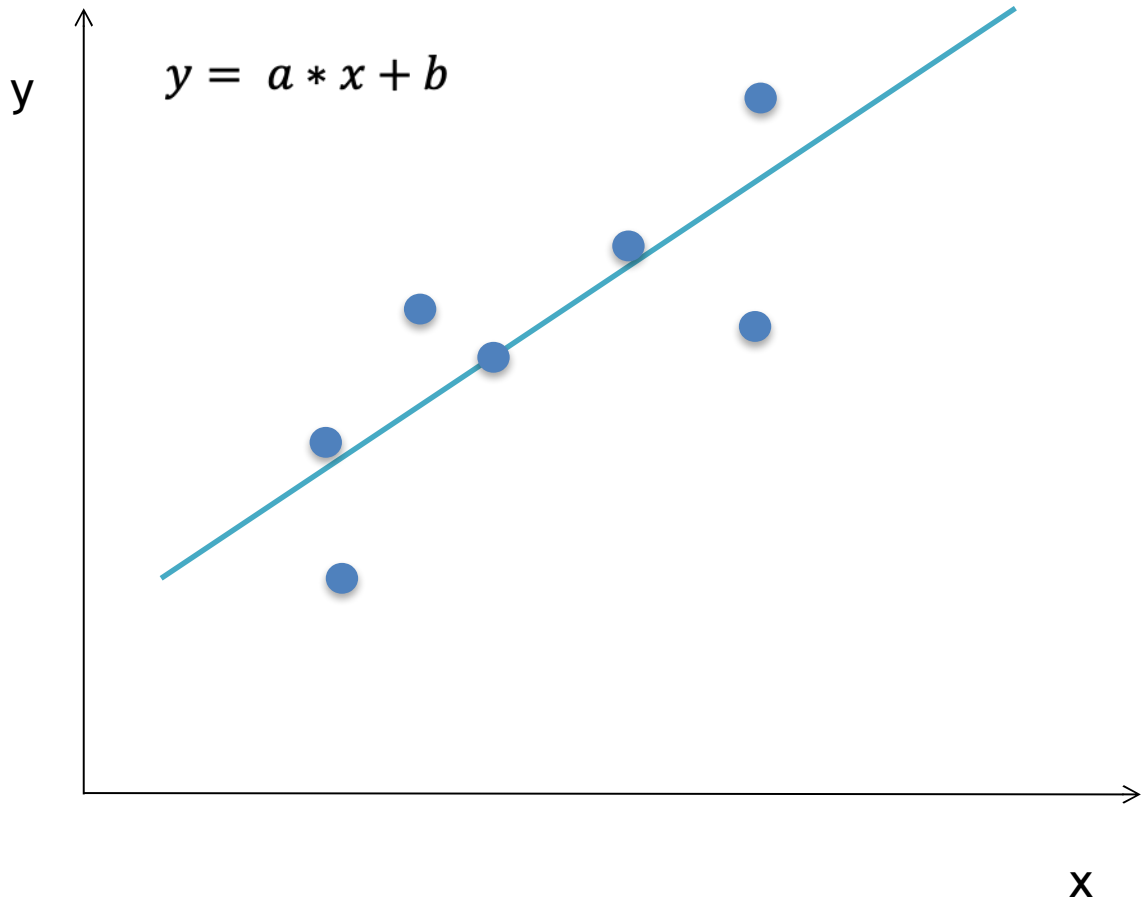

Price vs Square Footage (with regression line)
PRICE
500
1000
1500
2000
1000
1500
2000
2500
3000
3500
SQFT

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_baseme
0	7129300520	20141013T000000	221900.00	3	1.00	1180	5650	1.00	0	0	...	7	1180	0
1	6414100192	20141209T000000	538000.00	3	2.25	2570	7242	2.00	0	0	...	7	2170	400
2	5631500400	20150225T000000	180000.00	2	1.00	770	10000	1.00	0	0	...	6	770	0
3	2487200875	20141209T000000	604000.00	4	3.00	1960	5000	1.00	0	0	...	7	1050	910
4	1954400510	20150218T000000	510000.00	3	2.00	1680	8080	1.00	0	0	...	8	1680	0

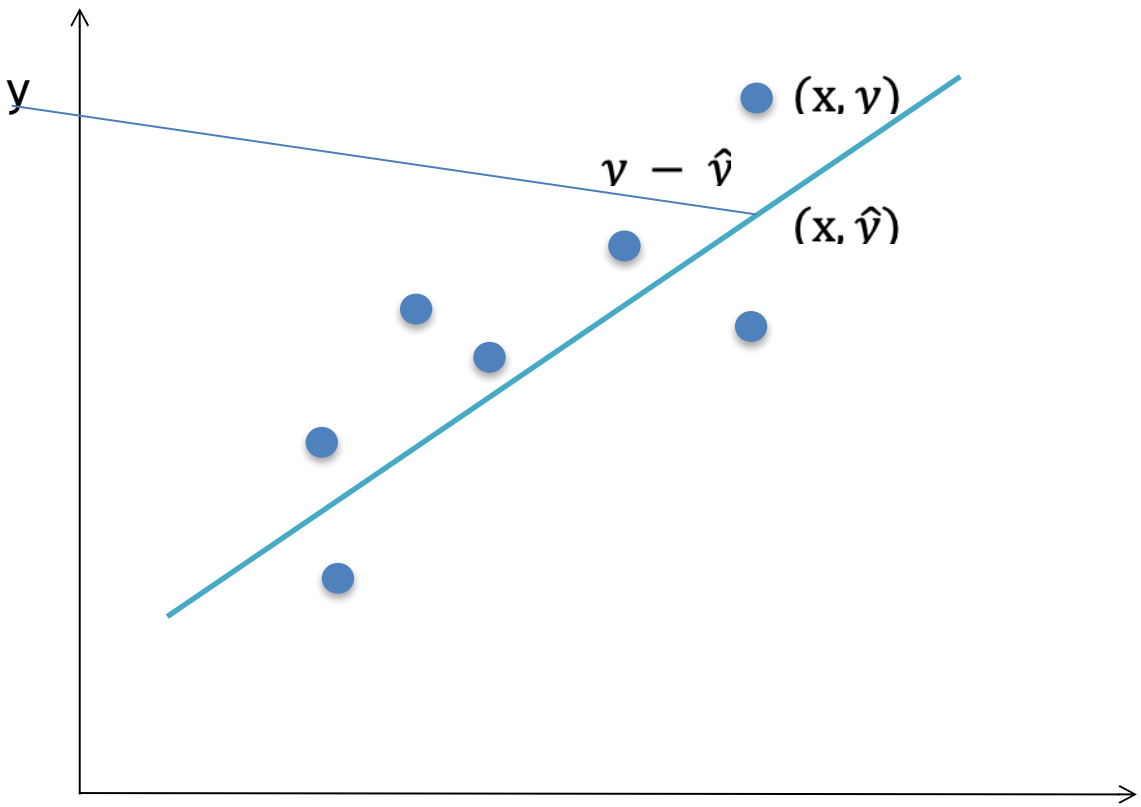
How to „fit” to data?

How to find the best prediction (regression curve)?

The best one, meaning the one with the smallest forecast error

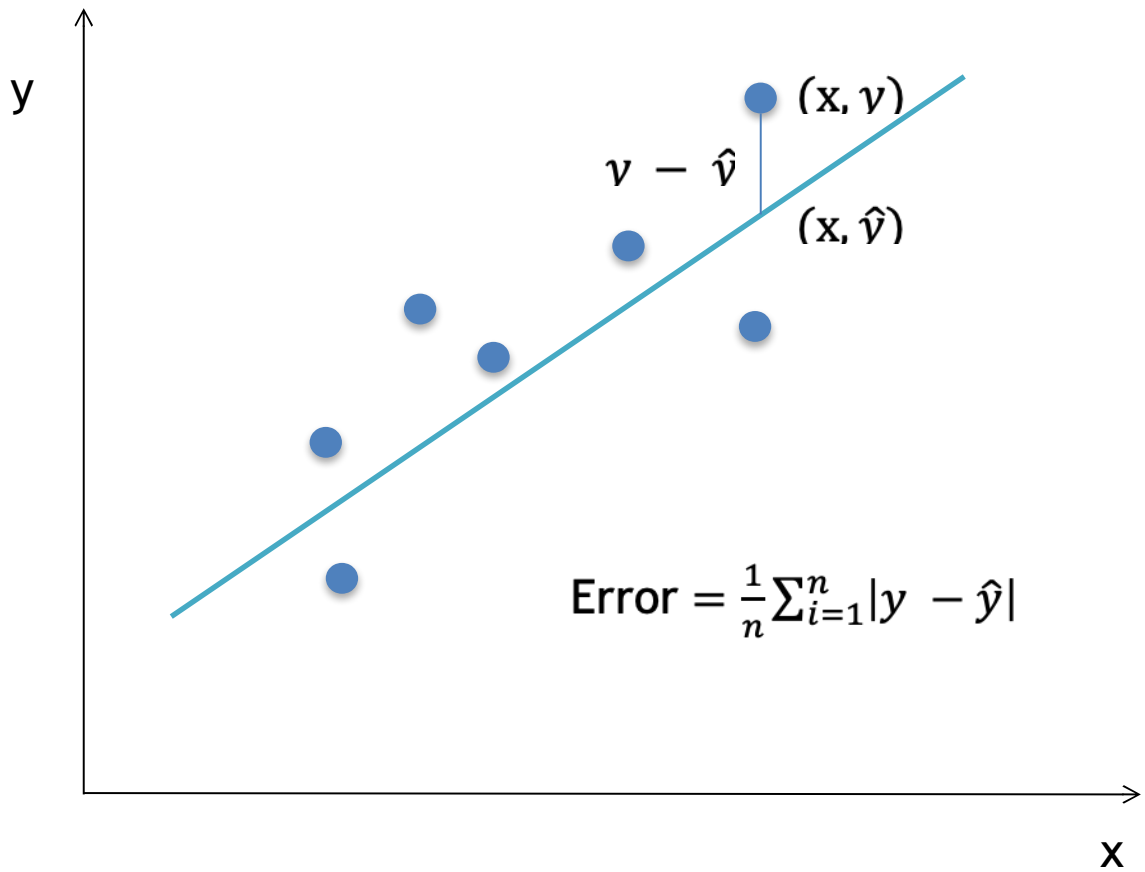


How do you measure forecast error?



Mean Absolute Error

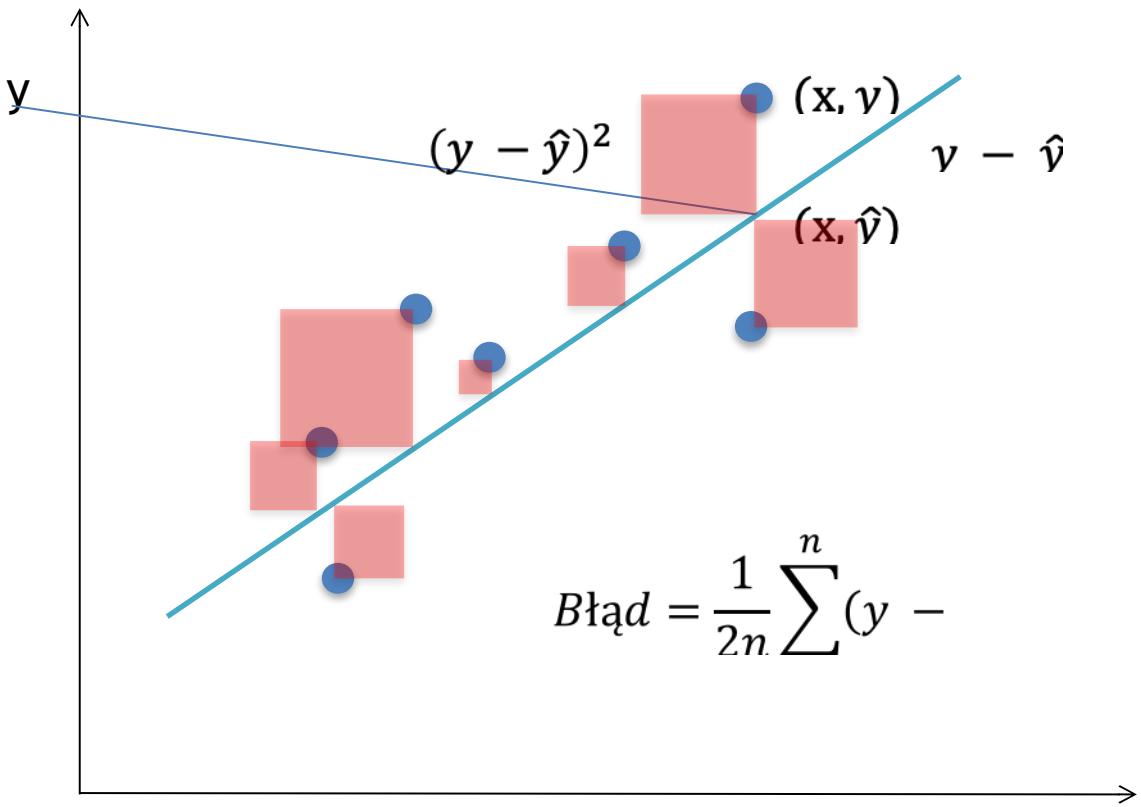
n = number of points



Mean Square Error

n = number of points

2 – for convenience

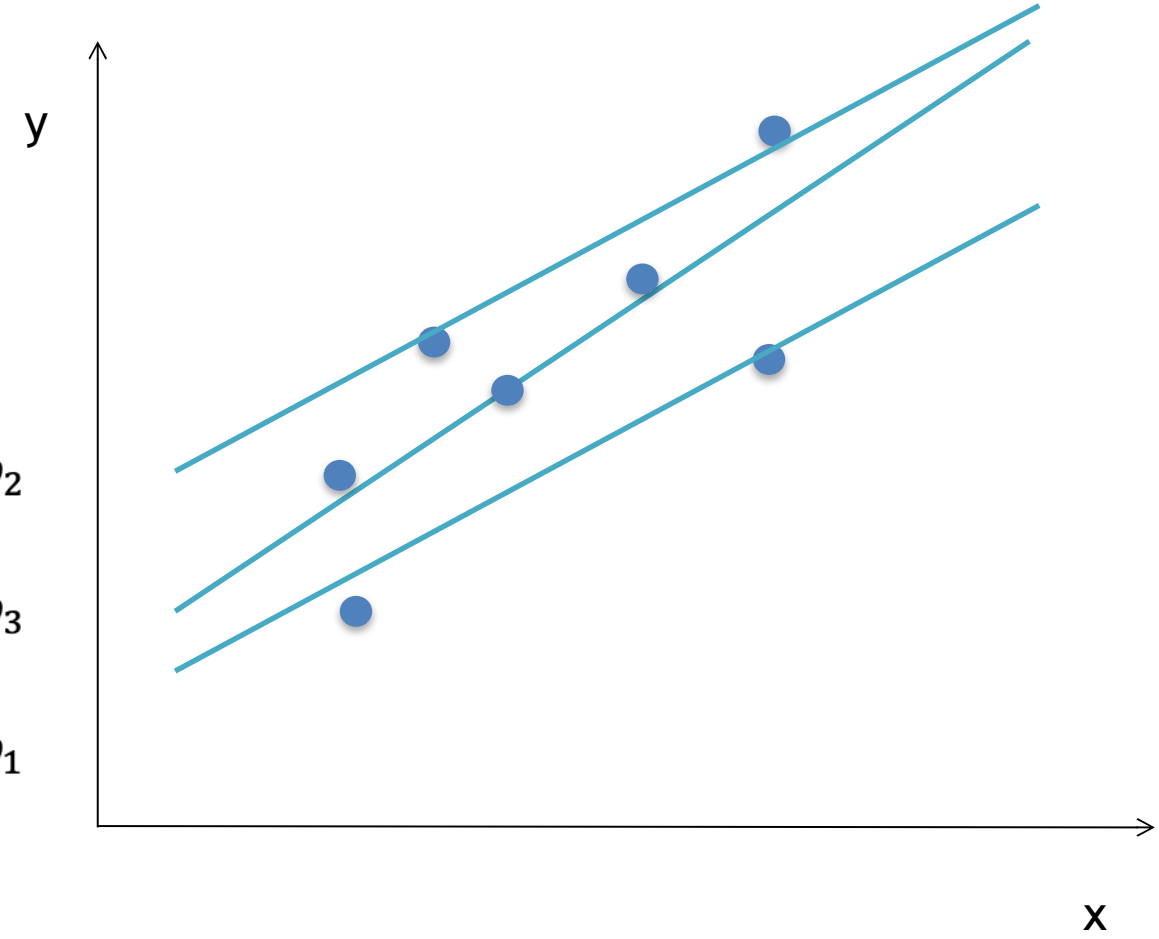


How do you find better and better approximations?

$$y = a_2 * x + b_2$$

$$y = a_3 * x + b_3$$

$$y = a_1 * x + b_1$$



How do we find successively better approximations of the coefficients of a line?

$$?: \; a_1 \rightarrow a_2 \rightarrow a_3$$

$$?: \; b_1 \rightarrow b_2 \rightarrow b_3$$

Idea

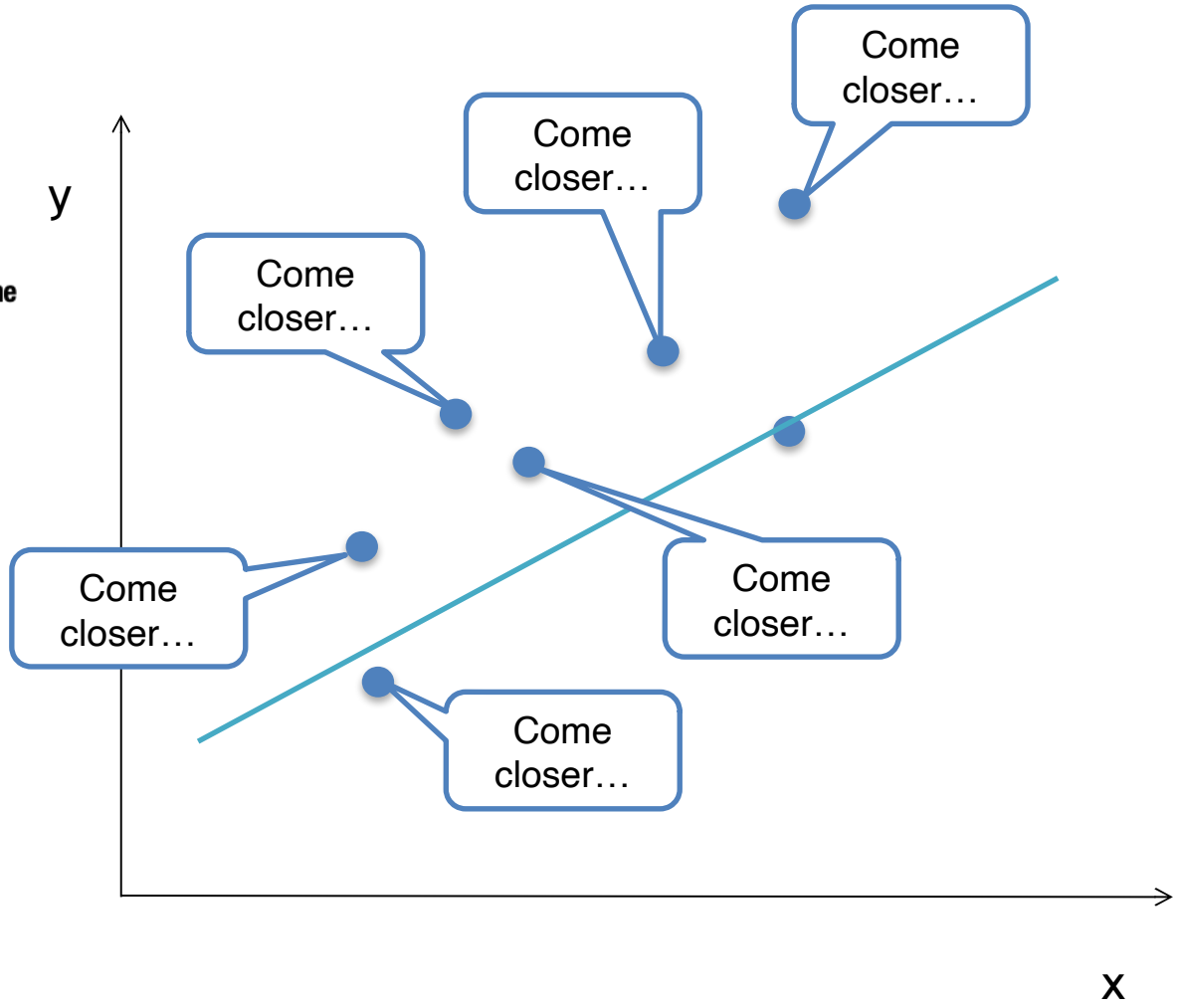
1. We start with random values of a_1 and b_1.

$$a_1 = Random$$
$$b_1 = Random$$

2. We "interrogate" each point: based on its recommendations, we change the coefficients of the current line :

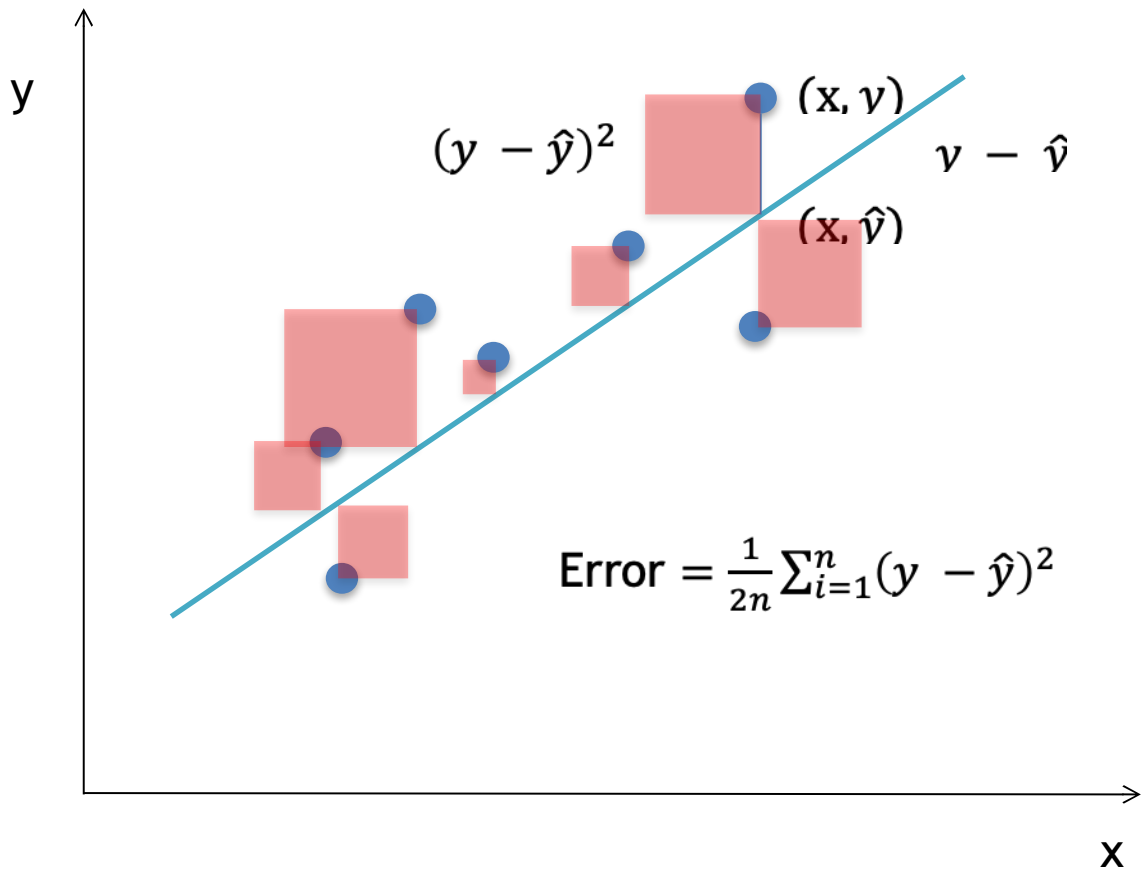
$$a_1 \rightarrow a_2,$$
$$b_1 \rightarrow b_2$$

3. For these changes we use the gradient descent method.

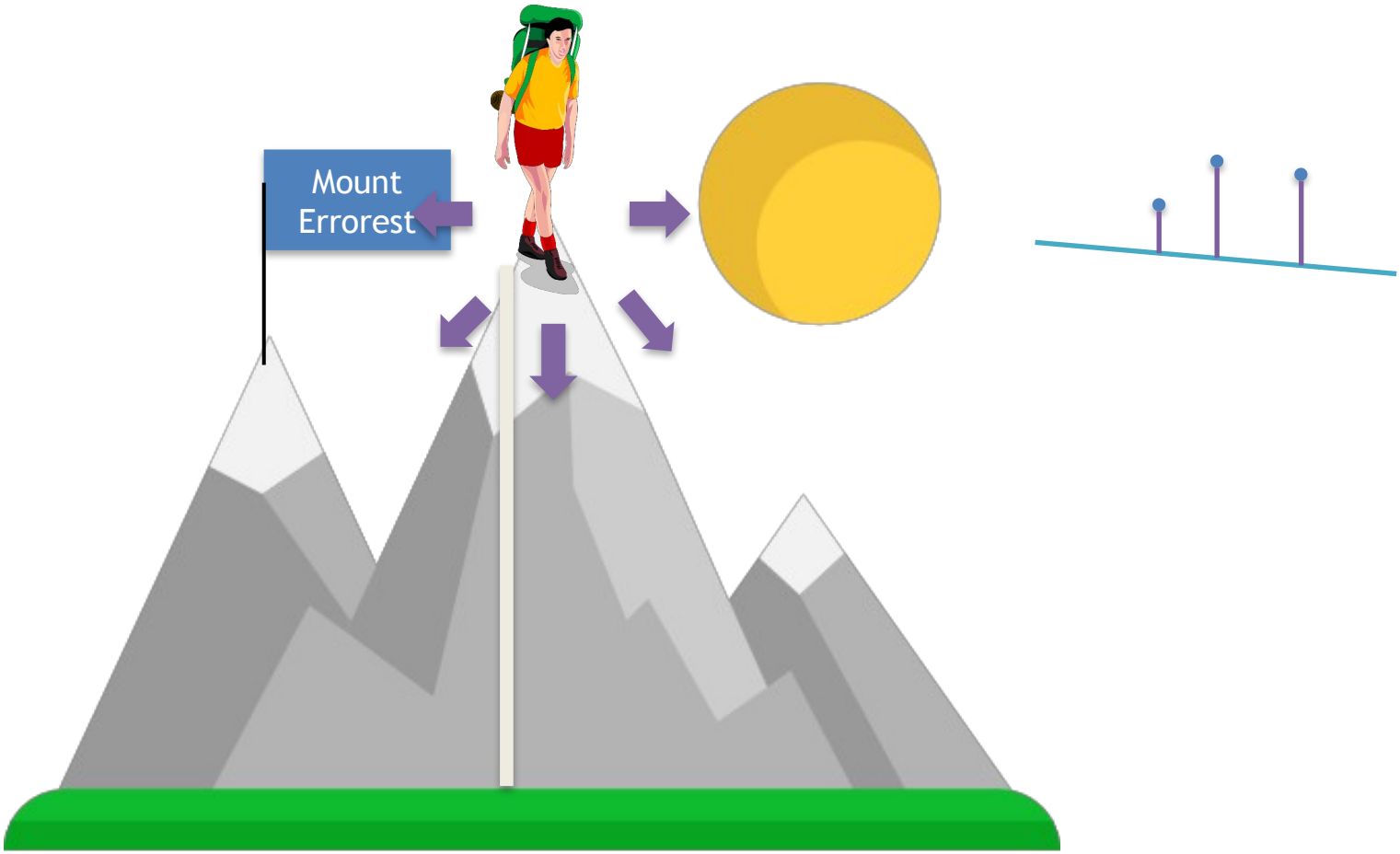


But how do you change a and b?

MSE mininimization by gradient descent method



Gradient descent



Gradient descent

Gradient descent

Gradient descent

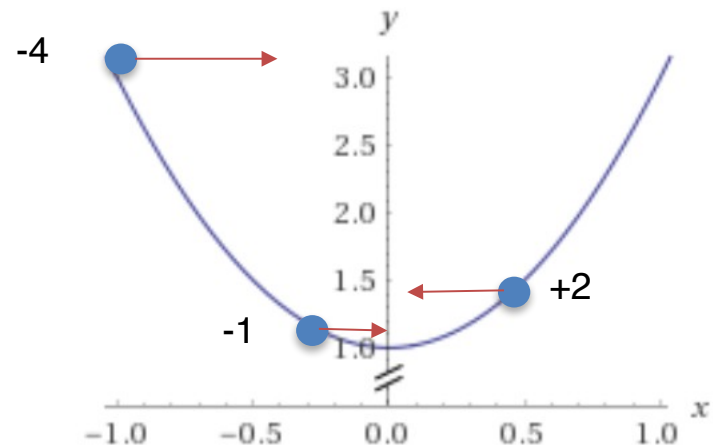
Gradient allows you to find the direction

Version one: change x :

$$x_1 \to x_1 - \left.\frac{dy}{dx}\right|_{x_1}$$

But this is too aggressive!
We need to be more subtle... So, we introduce the learning rate:

$$x_1 \to x_1 - \alpha \left.\frac{dy}{dx}\right|_{x_1}$$



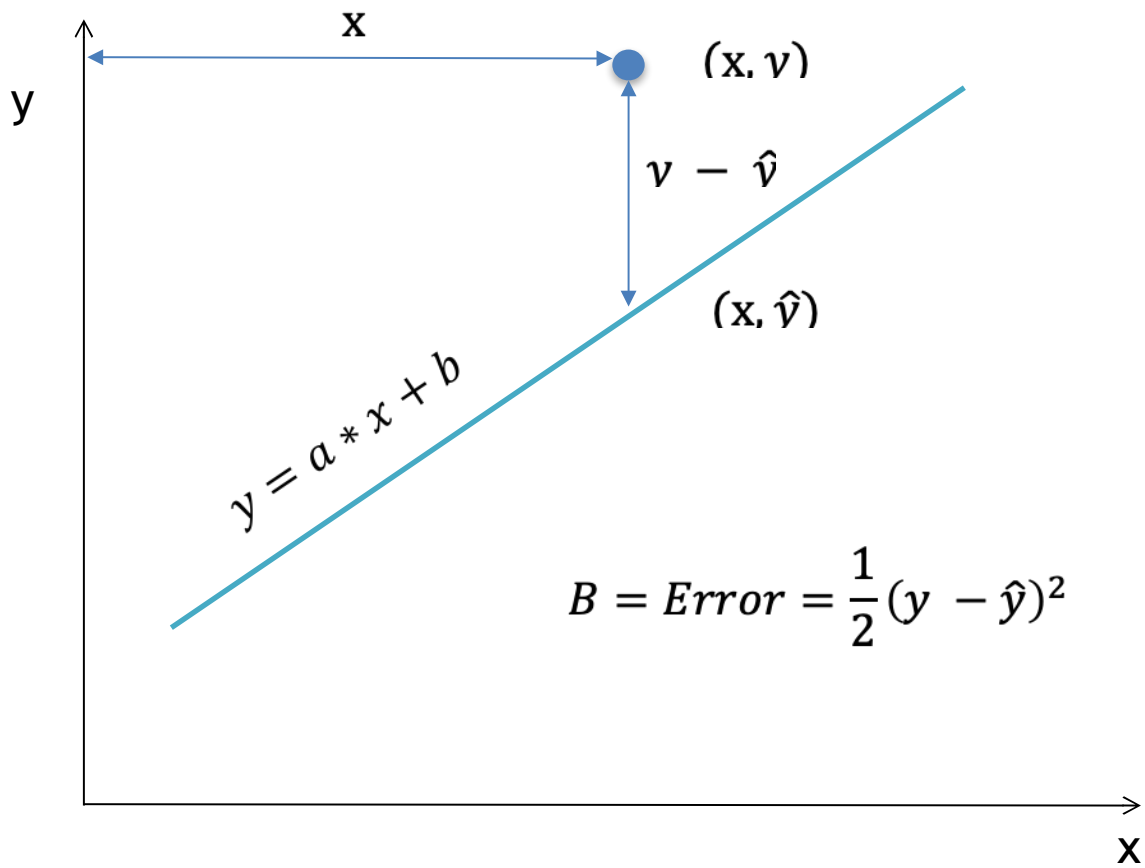
$$2x^2 + 1$$

$$\frac{d}{dx}(2x^2 + 1) = 4x$$

Gradient descent

Goal: Minimize MSEs

$$\frac{\partial B}{\partial a} = \frac{\partial}{\partial a}\left(\frac{1}{2}(y - \hat{y})^2\right) =$$
$$\frac{\partial}{\partial a}\left(\frac{1}{2}(y - a * x - b)^2\right) = ?$$



WolframAlpha computational intelligence.

d/da (1/2*(y - a*x - b)^2)

Bro

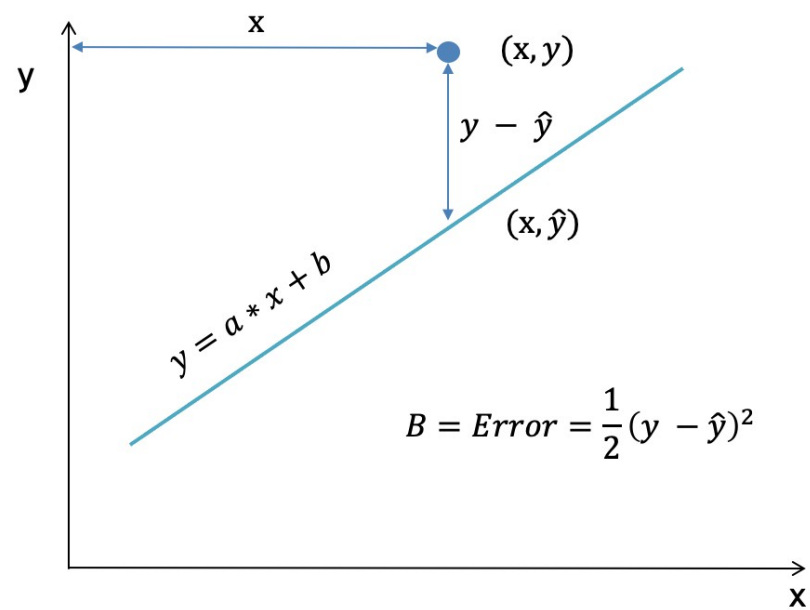
Derivative:

$$\frac{\partial}{\partial a}\left(\frac{1}{2}(y - a\,x - b)^2\right) = -x\,(-a\,x - b + y)$$

Iterations for a i b

$$\frac{\partial B}{\partial a} = \frac{\partial}{\partial a}\left(\frac{1}{2}(y - \hat{y})^2\right) = -(y - \hat{y})\mathrm{x}$$

$$\frac{\partial B}{\partial b} = \frac{\partial}{\partial b}\left(\frac{1}{2}(y - \hat{y})^2\right) = -(y - \hat{y})$$



Our step:

$$a \rightarrow a - \alpha\frac{\partial B}{\partial a} = a + \alpha(y - \hat{y})\mathrm{x}$$

$$b \rightarrow b - \alpha\frac{\partial B}{\partial b} = b + \alpha(y - \hat{y})$$

$$a \rightarrow a + \alpha(y - \hat{y})\mathrm{x}$$

$$b \rightarrow b + \alpha(y - \hat{y})$$

Stochastic gradient descent

1. We start with random values of a_1 and b_1.

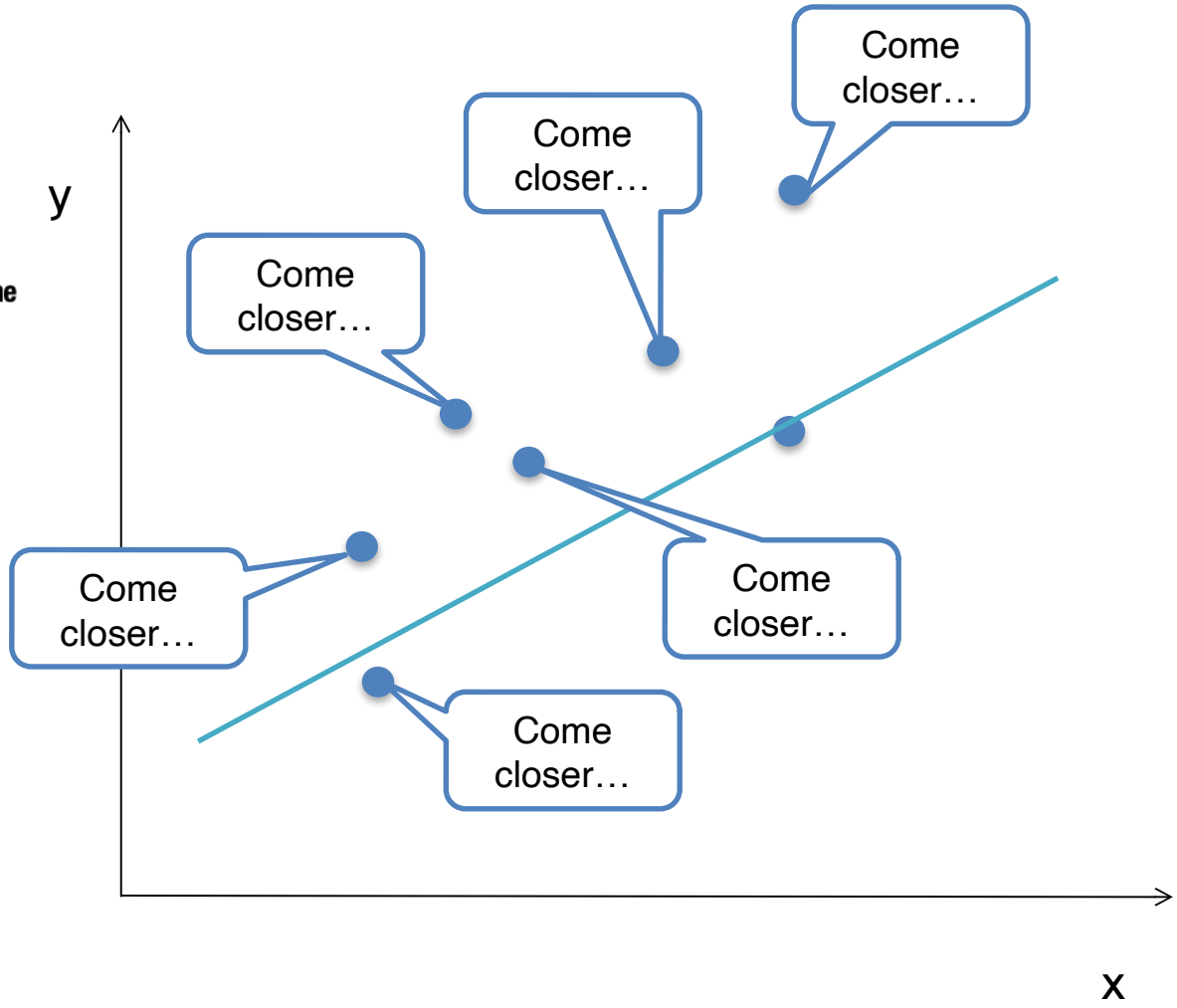
$$a_1 = Random$$
$$b_1 = Random$$

2. We „interrogate" the first point: based on its recommendations, we change the coefficients of the current straight line :

$$a_1 \rightarrow a_1 + \alpha(y - \hat{y})x,$$
$$b_1 \rightarrow b_1 + \alpha(y - \hat{y})$$

3. We interrogate another of the points
And we change the curve again...

4. In this way, we go through all the points



Batch gradient descent

1. **We start with random values of a_1 and b_1**

$$a_1 = Random$$
$$b_1 = Random$$

2. We "interrogate" the first of the points and identify the change in a and b based on its recommendations. We do not change the curve.

$$a_1 \rightarrow a_1 + \alpha(y - \hat{y})x = a_1 + \Delta a_{P1}$$
$$b_1 \rightarrow b_1 + \alpha(y - \hat{y}) = b_1 + \Delta b_{P1}$$

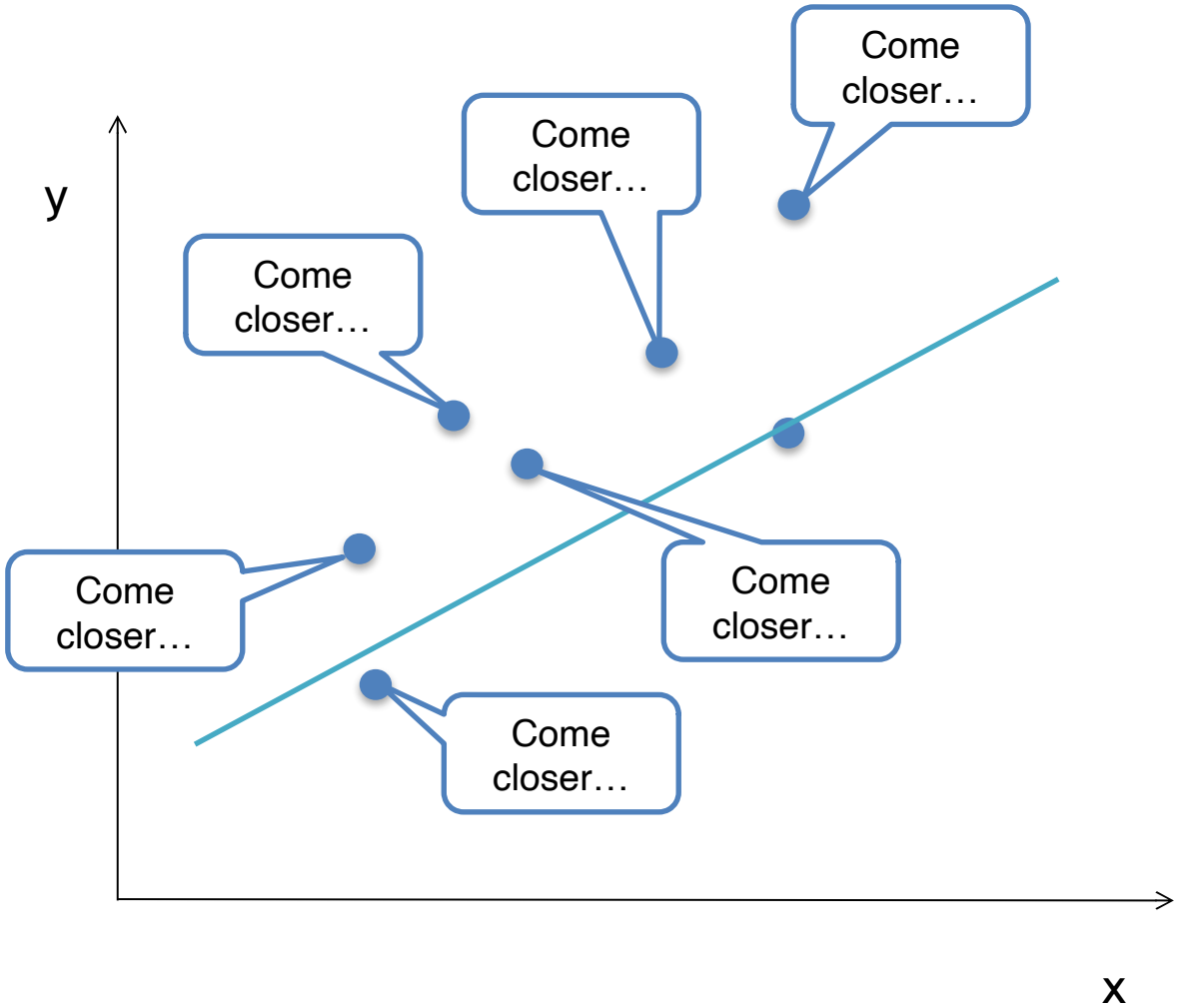
3. We listen to the next of the items and we record the change that this item recommends

$$a_1 \rightarrow a_1 + \alpha(y - \hat{y})x = a_1 + \Delta a_{P2}$$
$$b_1 \rightarrow b_1 + \alpha(y - \hat{y}) = b_1 + \Delta b_{P2}$$

4. In this way, we go through all the points

5. Finally, we change

$$a_1 \rightarrow a_1 + \overline{\Delta a_{Pi}}$$
$$b_1 \rightarrow b_1 + \overline{\Delta b_{Pi}}$$



Mini-batch gradient descent

1. We start with random values of a_1 and b_1.

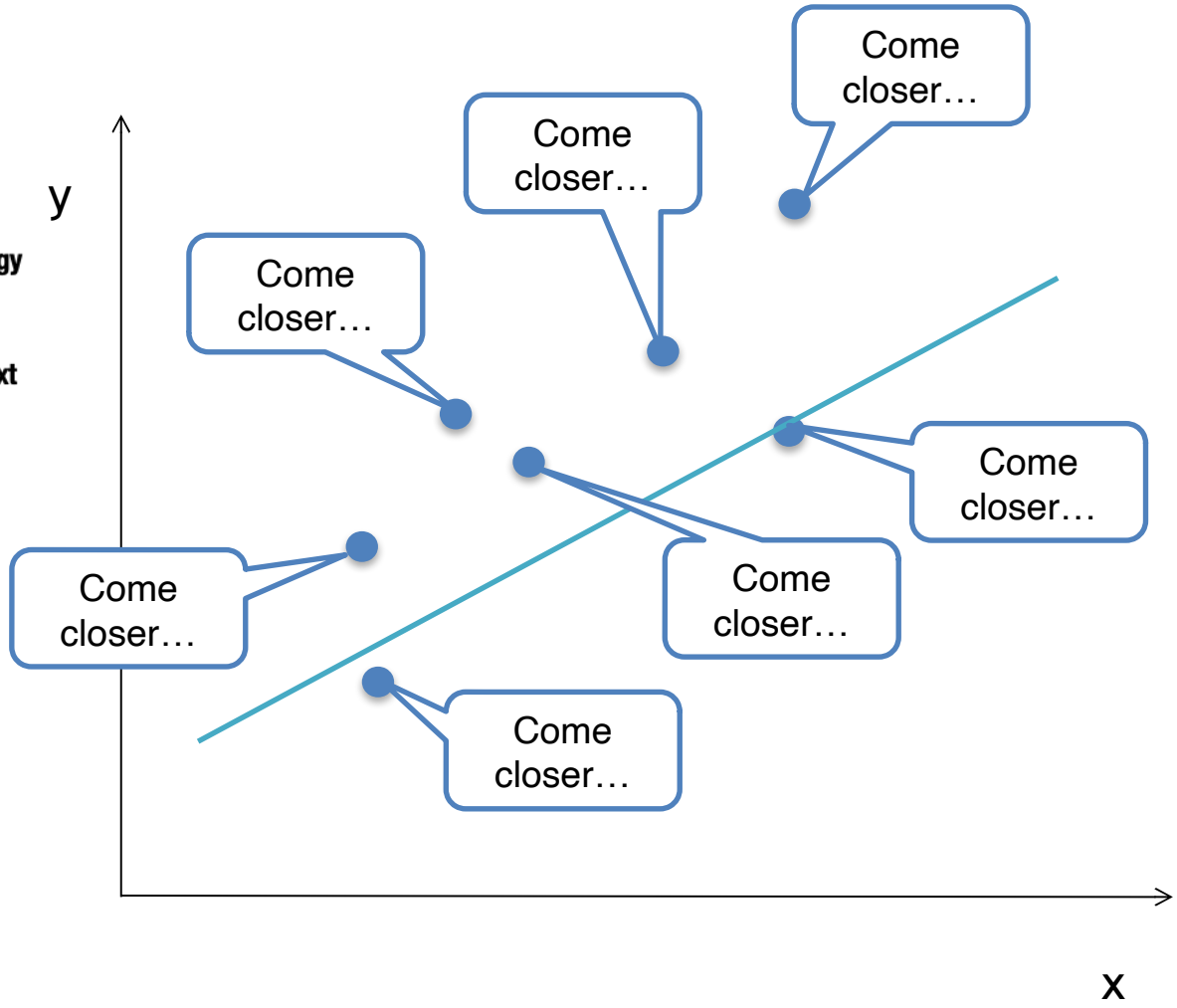
$$a_1 = Random$$
$$b_1 = Random$$

2. We implement the batch gradient descent strategy for the first group of points

3. We change the curve

4. We implement batch gradient descent for the next group of points

5. … until convergence

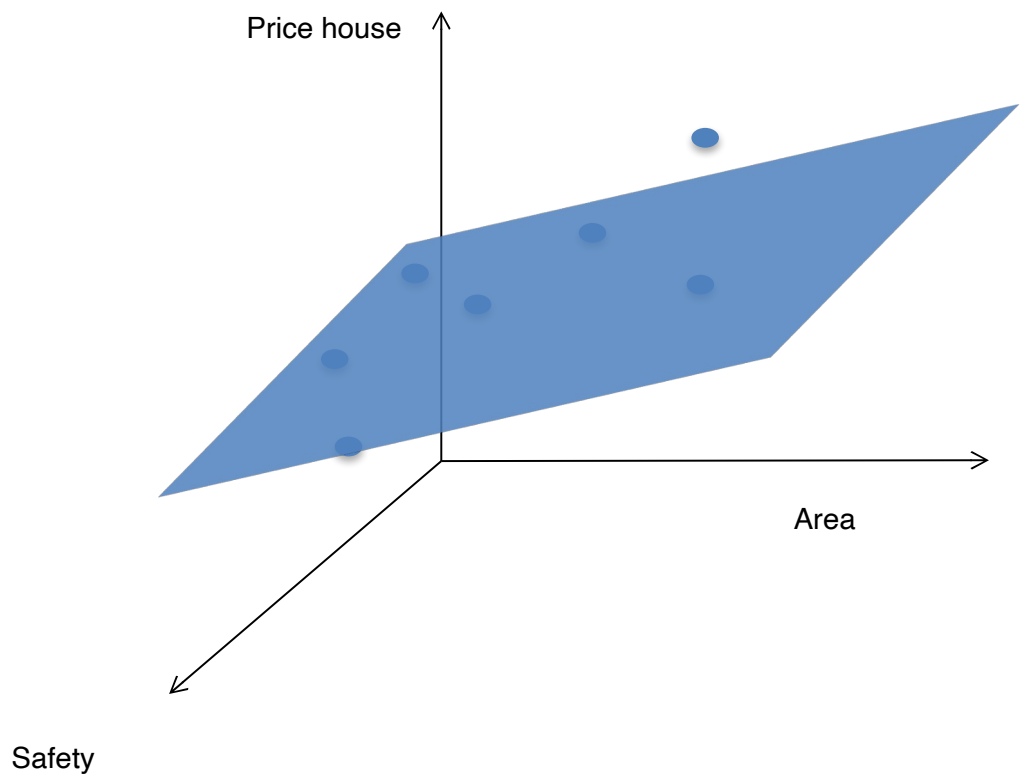


More dimensions

Regression effect: plane (not line)

$$\text{Price} = w_1 * (safety) + \\ + w_2 * (area) + w_3$$

Procedure: exactly the same (gradient descent), but for 3 variables



More dimensions

For an n - dimensional space (n-1 explanatory variables)
the solution to the regression problem is the (n-1) dimensional hyperplane equation

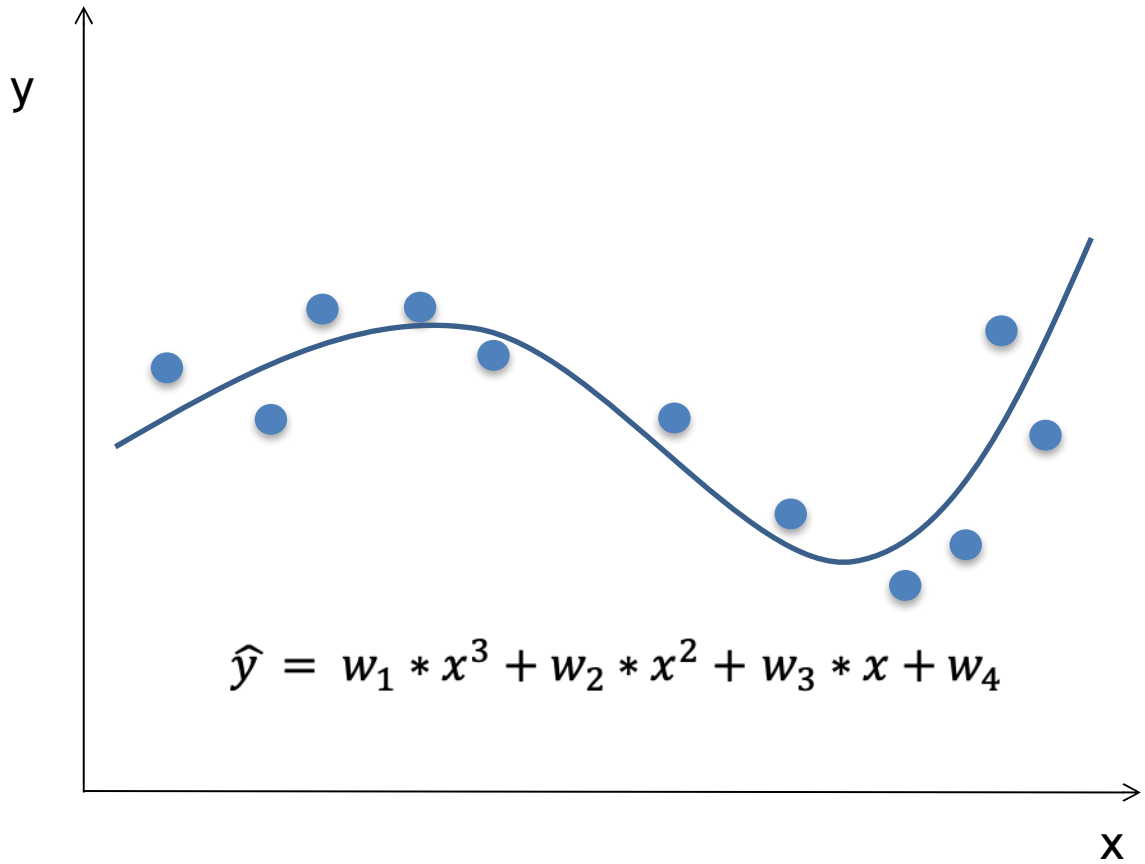
$$\hat{y} = w_1 * x_1 + w_2 * x_2 + \ldots w_{n-1} * x_{n-1} + w_n$$

Procedure: exactly the same (gradient descent), except that for n variables

	x_1 Arae	x_1 Education quality	x_1 Safety	...	$\hat{y}$ Price
House 1	150	5	5		800 000
House 2	50	3	2		300 000
House 3	250	4	4		900 000

Polynomial regression

Procedure: exactly the same (gradient descent)



$$\hat{y} = w_1 * x^3 + w_2 * x^2 + w_3 * x + w_4$$